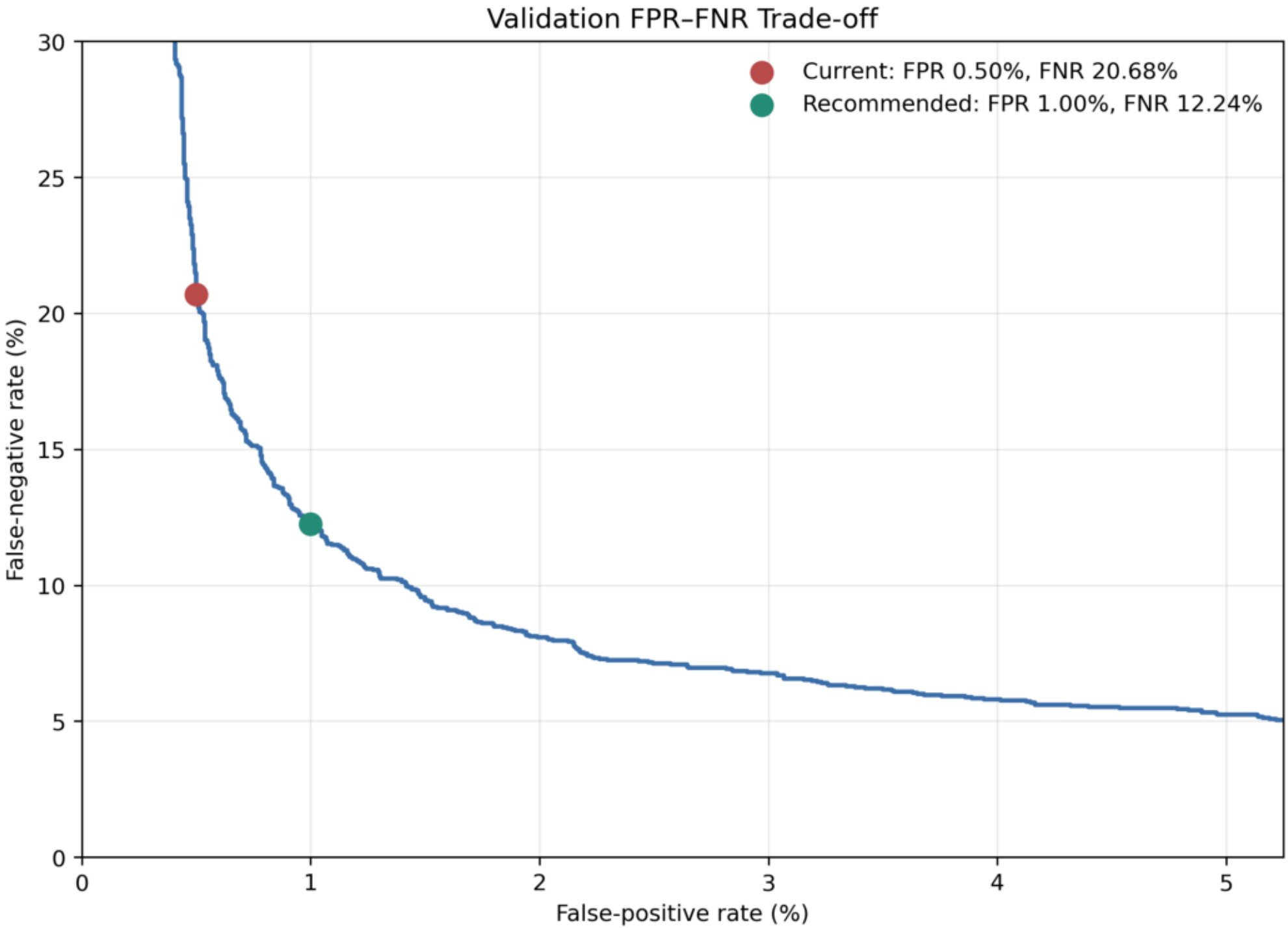
Calibration rows: 23,100 (20,600 legitimate, 2,500 phishing)
Current: threshold 0.995307744, FPR 0.50%, FNR 20.68%, F1 86.48%
Recommended: threshold 0.970766783, FPR 1.00%, FNR 12.24%, F1 89.55%
Validation FNR reduction: 8.44 percentage points

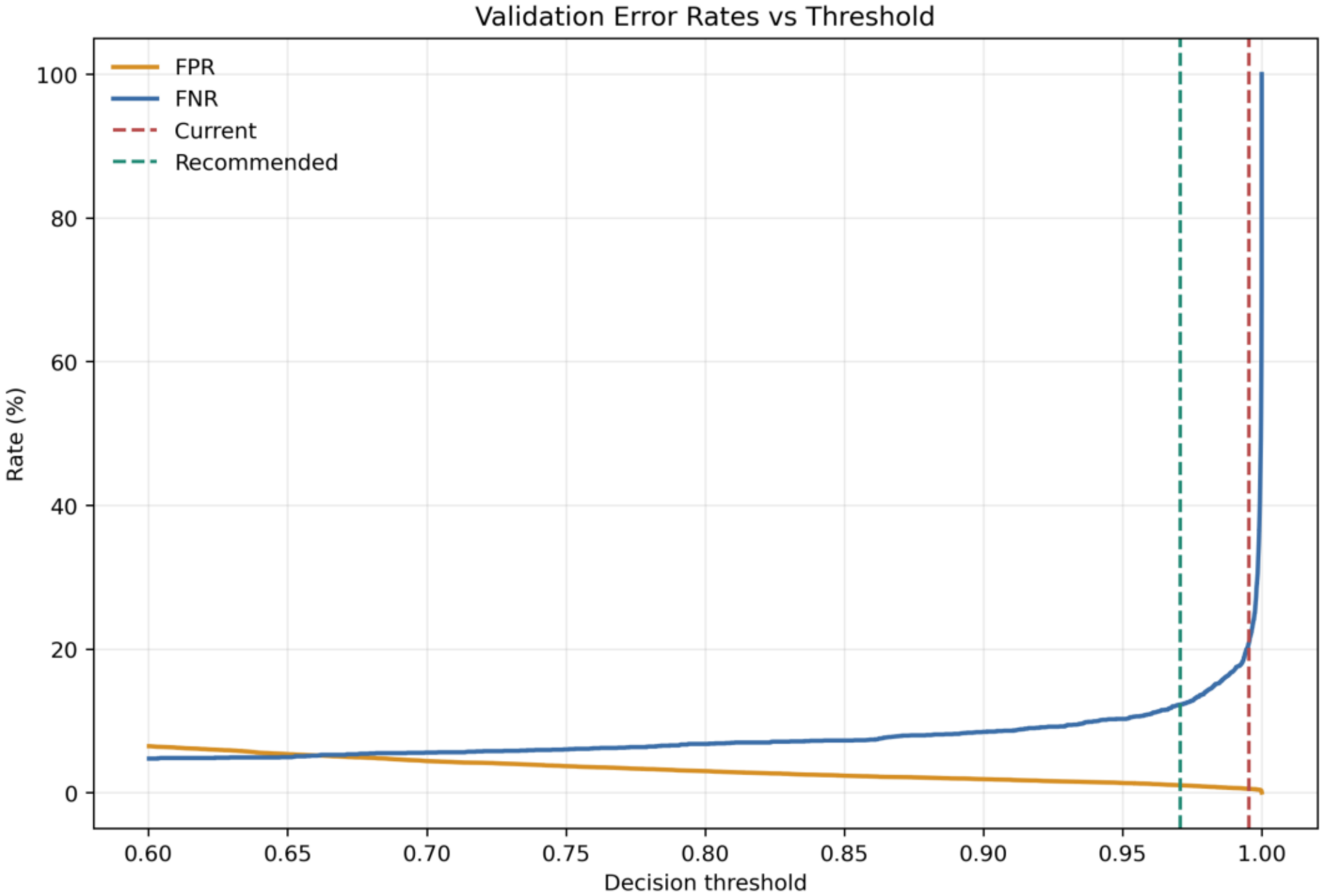
FPR/FNR operating-point analysis

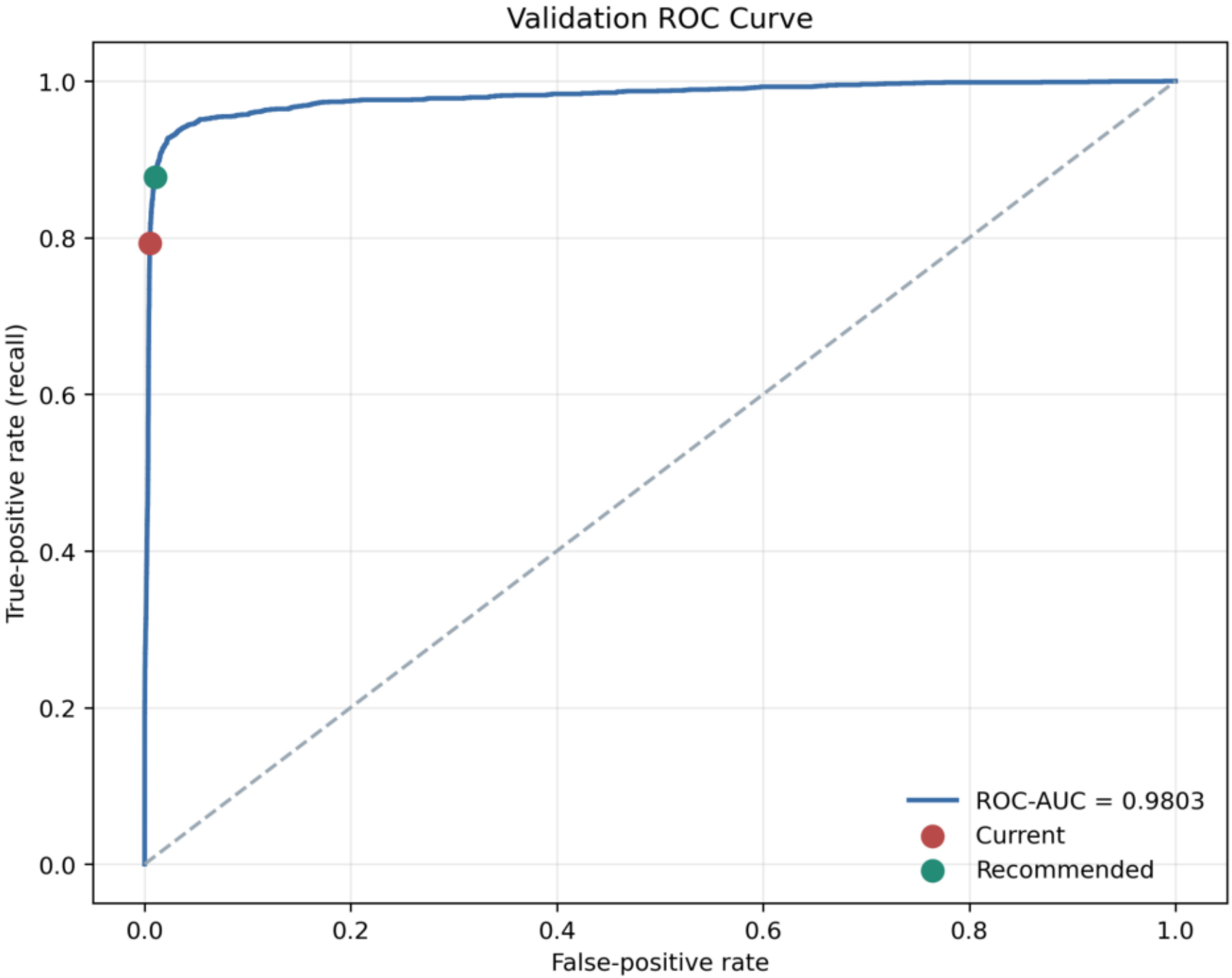
HYB-MS +10% MS-HardGAN-C browser ensemble · validation-only selection

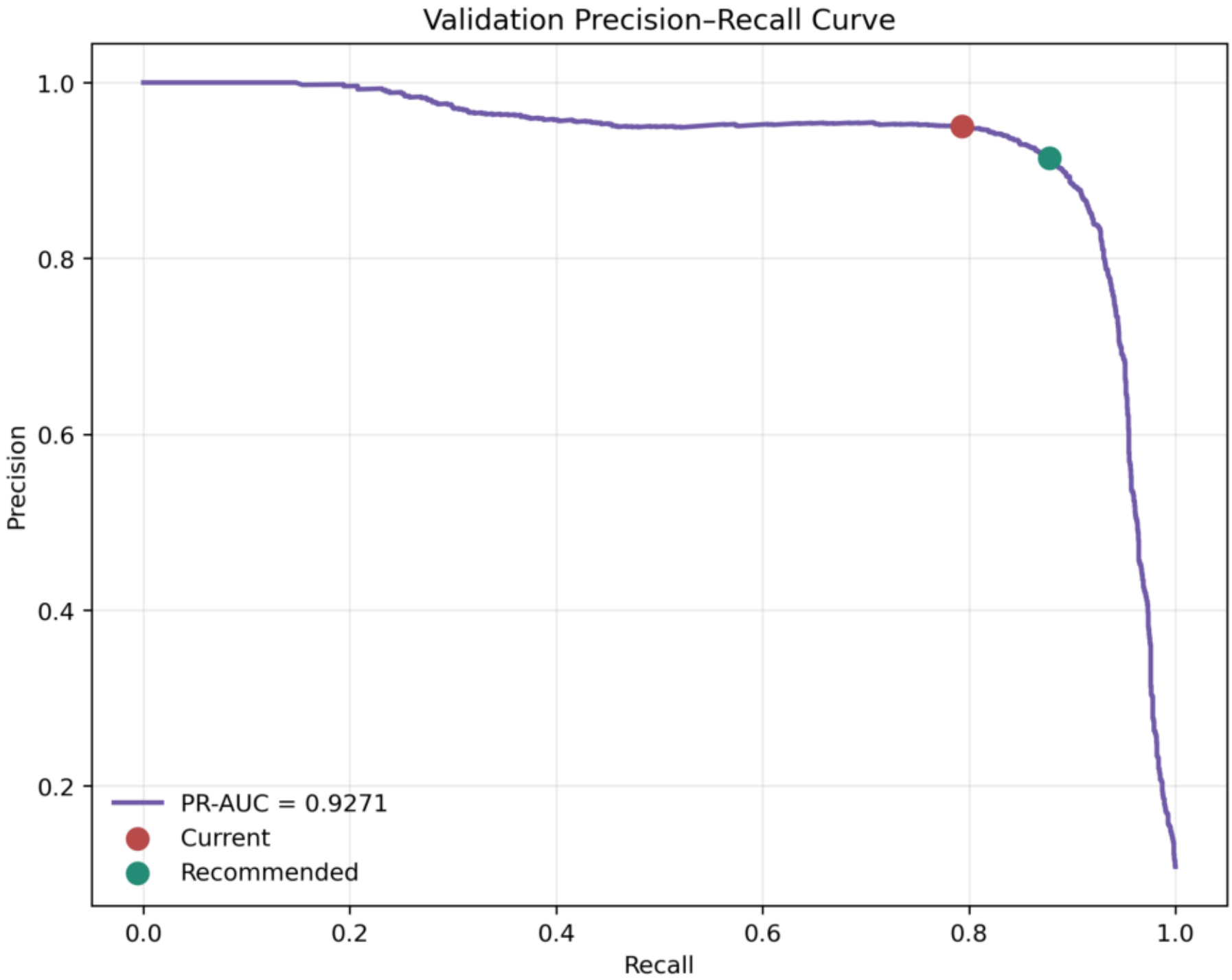
Recommendation: use the 1% validation-FPR operating point for a recall-oriented passive warning.
Do not lower the automatic block threshold to this value.

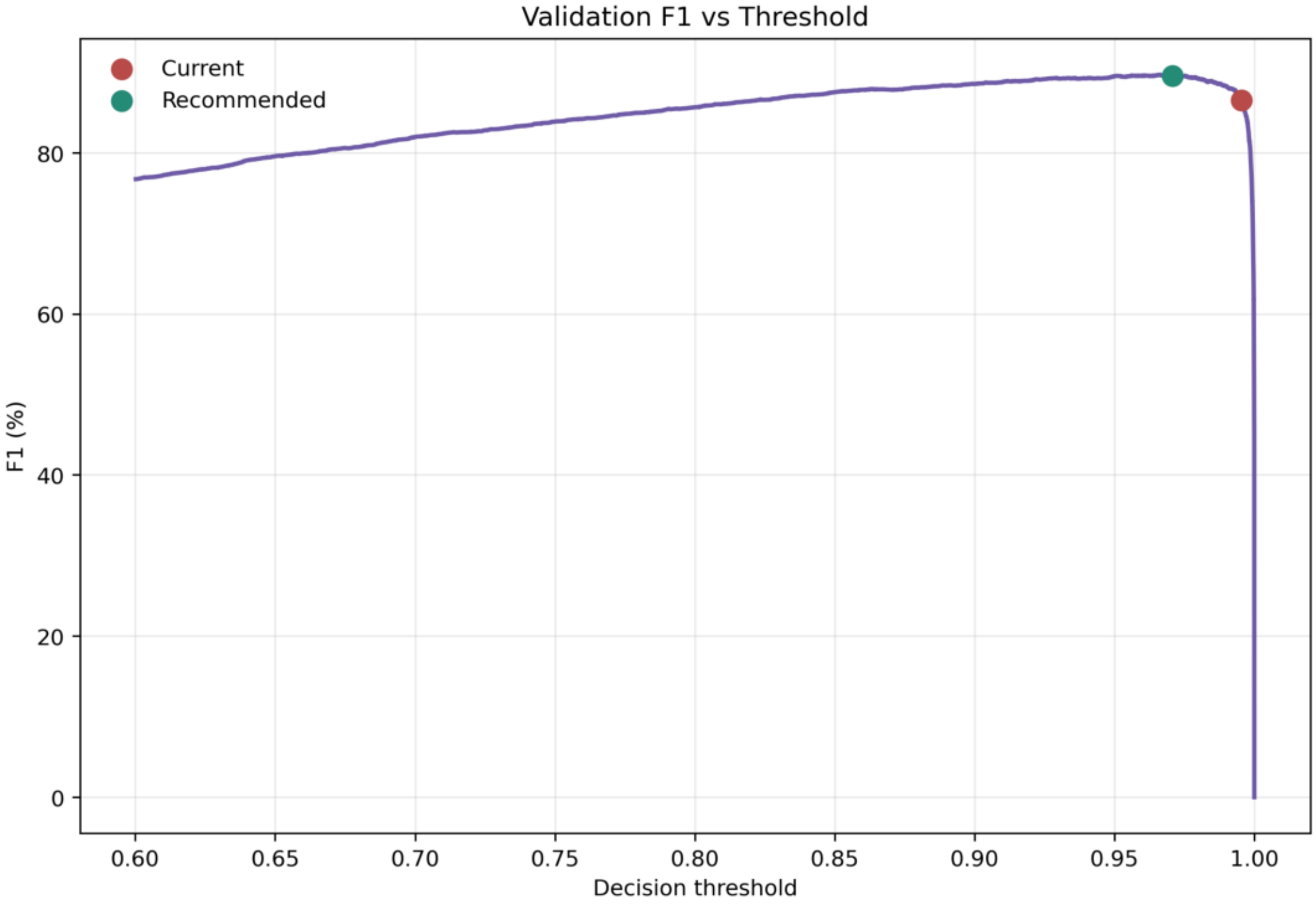
Thresholds were selected without using frozen-test labels. The 10K test set was used only after selection to report confirmation behavior.

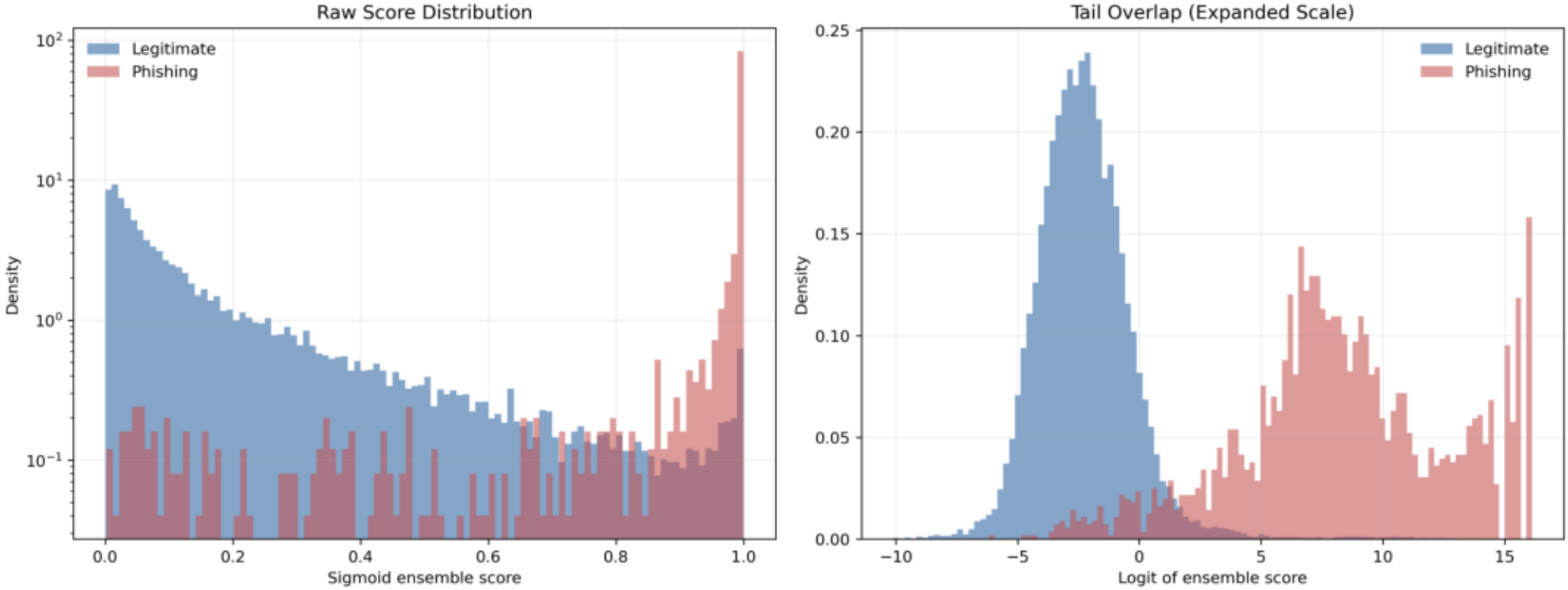

Validation FPR–FNR Trade-off
Current: FPR 0.50%, FNR 20.68%
Recommended: FPR 1.00%, FNR 12.24%
False-negative rate (%)
False-positive rate (%)
0
5
10
15
20
25
30
0
1
2
3
4
5


Validation Error Rates vs Threshold
FPR
FNR
Current
Recommended
100
80
60
40
20
0
Rate (%)
0.60
0.65
0.70
0.75
0.80
0.85
0.90
0.95
1.00
Decision threshold


Validation ROC Curve
True-positive rate (recall)
False-positive rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC-AUC = 0.9803
Current
Recommended


Validation Precision–Recall Curve
Precision
Recall
1.0
0.8
0.6
0.4
0.2
0.0
0.2
0.4
0.6
0.8
1.0
PR-AUC = 0.9271
Current
Recommended


Validation F1 vs Threshold
Current
Recommended
F1 (%)
0
20
40
60
80
0.60
0.65
0.70
0.75
0.80
0.85
0.90
0.95
1.00
Decision threshold


Validation Score Separation
Raw Score Distribution
Legitimate
Phishing
Density
Sigmoid ensemble score
Tail Overlap (Expanded Scale)
Legitimate
Phishing
Density
Logit of ensemble score